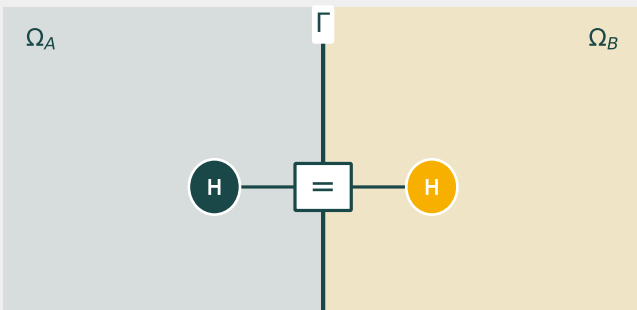


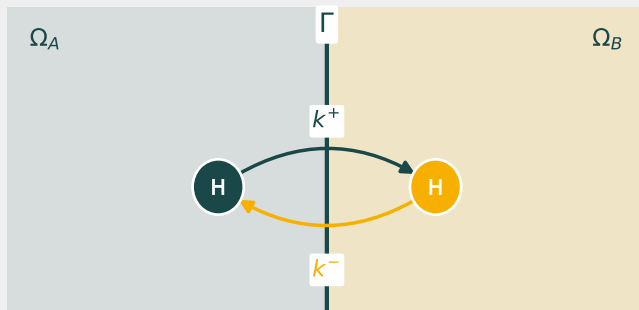
(a) local thermodynamic equilibrium



$$c_A/K_{S,A} = c_B/K_{S,B}$$

no rate constant, no timescale, no resistance

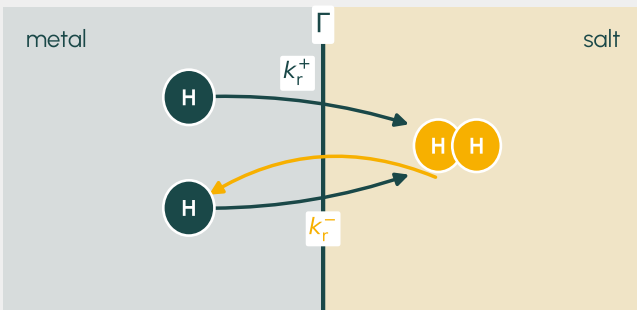
(b) Model 1: first-order exchange



$$\phi = k^+ c_A|_{\Gamma} - k^- c_B|_{\Gamma}$$

$$k^+/k^- = K_{S,B}/K_{S,A}, \text{ LTE as } k^{\pm} \rightarrow \infty$$

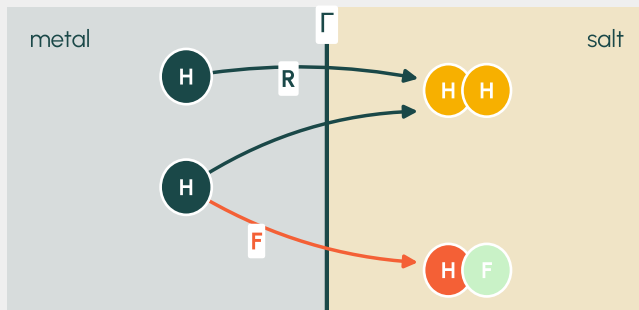
(c) Model 2: recombination into a carrier



$$w_{\text{rec}} = k_r^+ (c_H^m)^2 - k_r^- c_{H_2}^s$$

atoms leave at $2w_{\text{rec}}$, one molecule enters at w_{rec}

(d) Model 3: competing channels



$$R: 2H \rightleftharpoons H_2 \text{ (quadratic)} \quad F: H + a_F \rightleftharpoons HF \text{ (linear)}$$

$\mathcal{B} = w_F/2w_{\text{rec}}$ sets the split, and it is kinetic